

Revising the SETI Post-Detection Protocols for the 2020s and Beyond: A Report on Work in Progress

Carol Oliver^a*, Kathryn Denning^b, Leslie I. Tennen^c, Michael A. Garrett^d

^a Australian Centre for Astrobiology, School of Biological, Earth, and Environmental Sciences, University of New South Wales, Australia, carol.oliver@unsw.edu.au

^b Dept of Anthropology, York University, Toronto, Canada, kdenning@yorku.ca

^c Law Offices of Sterns and Tennen, Phoenix, Arizona, USA. ltennen@astrolaw.com

^d Sir Bernard Lovell Chair of Astrophysics, University of Manchester, Director of the Jodrell Bank Centre for Astrophysics, United Kingdom, michael.garrett@manchester.ac.uk

* Corresponding Author

Abstract

In 1989, the International Academy of Astronautics (IAA) adopted the “Declaration of Principles Concerning Activities Following the Detection of Extraterrestrial Intelligence”, also known as the Search for Extraterrestrial Intelligence (SETI) Post-Detection Protocols. These provided a set of guiding principles to be followed in the event of a putative signal detection, having been developed by the IAA SETI Permanent Committee over a period of several years starting in the late 1980s, and subsequently approved by the International Institute of Space Law (IISL), and endorsed by the Committee on Space Research (COSPAR), the International Astronomical Union (IAU), and the (International Astronautical Federation (IAF). A streamlined update to reflect the emergence of the Internet was developed and accepted by the IAA SETI Permanent Committee in 2010 as the “Declaration of Principles Concerning the Conduct of the Search for Extraterrestrial Intelligence”.

Since then, the nature of global communication has been dramatically changed by social media, and multiple SETI ‘false alarms’ have demonstrated the ways in which information moves in our present era. Simultaneously, SETI searches – now also known techno-signature searches – have radically expanded in extent, scope, and power, with substantial search efforts underway at Breakthrough Listen, the SETI Institute, the Chinese Five-hundred-meter Aperture Spherical Telescope (FAST), and many other observatories around the world, between them using many of the world’s most powerful telescopes. Moreover, upcoming next-generation telescopes will allow the detection of molecules indicative of the existence of life in exoplanet atmospheres, conduct high cadence all-sky surveys and be able to detect the equivalent of an airport radar 100 light years away. As a result, the community of scientists engaged in SETI research is expanding significantly. In addition, other methods of detecting exoplanets as potential techno-signature targets are increasing, broadening the community of scientists who may not regard themselves as SETI-related but could potentially uncover data of SETI interest.

In 2022, the IAA SETI Committee decided that a high priority was to update the 2010 IAA Declaration of Principles. A proposal for a small task group with the sole task of updating these Principles was initiated at the end of 2022, and co-chairs put in place to develop that idea. Given rapid changes in SETI/technosignatures research and the dynamic communications landscape, we now envision an eventual combination of timeless Principles which generally preserve the original intentions of the 1989 Protocols and Codes of Conduct and Best Practices documents to provide additional regularly updated guidelines, responsive to changing circumstances. We present a report on the progress on the formation of the task group and its deliberations.

Keywords: SETI, techno-signatures, Protocols, post-detection, discovery, Declaration of Principles

1. Introduction

The IAA SETI Protocols are comprised of a series of three documents, prepared over a period of more than 20 years by the IAA committee, however denominated, dedicated to SETI. (“SETI Committee” is used herein as a generic name for this committee in its various permutations) [1] [2]. The first document, the Declaration of Principles Concerning Activities Following the Detection of ExtraTerrestrial (sic)

Intelligence (Post-Detection Protocols) [3] was released in 1989. It was followed in 1995 by the Draft Declaration of Principles Concerning Sending Communications with Extraterrestrial Intelligence (Reply Protocols) [4]. The draft Reply Protocols were not further formalised, but the Post-Detection Protocols were updated in 2010 in the Declaration of Principles Concerning the Conduct of the Search for Extraterrestrial Intelligence (Declaration of Principles).

[5]. ("Protocols" is used herein as a generic name for the three documents.)

The Post-Detection Protocols were prepared with specific reference to the search for radio signals, which was the predominant medium for conducting SETI at the time and did not anticipate the advent of a ubiquitous Internet, or social media. The 2010 update was also crafted primarily with radio SETI in mind. Updates are now needed due to concerns including (but not limited to): the very fast and complex communications environment which poses new challenges for scientists; the expansion of technosignatures searches; the potential for ambiguity or overlap between biosignatures and technosignatures, e.g., in exoplanet atmospheres; and the potential for discovery/confirmation processes more prolonged than the detection of an incoming radio transmission. N.b. the SETI Committee sees Unidentified Aerial Phenomena (UAPs) as being generally outside of its remit, therefore not included in either the 1989 SETI Protocols or the 2010 Declaration of Principles, and not considered either in the updating of the latter document.

Members of the IAA SETI Committee are diverse in terms of academic specialty and include representatives of institutions and organisations as well as unaffiliated practitioners. The SETI Committee meetings and symposia held in conjunction with the IAC attract considerably more international participation than any other entity, public or private. The SETI Committee holds a unique position within the global SETI community as the primary and pre-eminent multi-disciplinary platform for individuals actively engaged in all aspects of SETI to connect with each other and discuss their own methods, results, and dilemmas. Since the 1989 Protocols and the 2010 update were developed under the umbrella of the SETI Committee, it falls to the Committee to develop and achieve widespread acceptance of protocols to guide the global SETI community in the process of confirmation and announcement of the discovery of ETI (extraterrestrial intelligence). This includes the framework for guidance decisions relating to possible responses to any communicative evidence detected, and to establish recommendations for the protection of scientific integrity and the safety of individuals.

2. Background

In the mid-1980's, a consensus began to coalesce within the IAA SETI Committee that there was a need for an agreed-upon set of written principles to guide the international SETI community in maintaining the highest standards of scientific responsibility and credibility regarding dissemination of information concerning the detection of extraterrestrial intelligence. Many individuals were involved in this endeavour, most notably Allan E. Goodman, Michael A. G. Michaud,

and Jill Tarter. Goodman proposed the preparation of a post-detection protocol during the 15th Review Meeting on the Search for Extraterrestrial Intelligence, held during the 37th International Astronautical Federation Congress in Innsbruck in 1986, and subsequently revised in [6] "Diplomatic and Political Problems Affecting the Formulation and Implementation of an International Protocol for Activities Following the Detection of a Signal from Extraterrestrial Intelligence".

Goodman's proposed Protocol, "based on interviews with scientists and national space policy makers", was comprised of three articles:

"1. Any individual, public, or private astronomical observatory, or governmental authority which has detected a signal from, or other evidence of extraterrestrial intelligence will freely and openly announce their findings and submit all data necessary for independent verification to the international scientific community.

"2. Plans for analysing the evidence of an extraterrestrial signal and any response to an actual communication form [sic] an extraterrestrial intelligence will be formulated by a Standing Consultative Commission chartered under the auspices of the International Astronautical Union [sic].

"3. In the event that a signal from extraterrestrial intelligence appears to pose a threat to human health or peace, nations will immediately consult the Security Council of the United Nations."

The following year the 38th IAF Congress in Brighton, England held a session entitled "Formulation of an International Protocol for Activities Following the Detection of a Signal from Extraterrestrial Intelligence" which culminated in a drafting session open to all interested parties led by Michaud [7]. After several rounds of comment and revision, the Post-Detection Protocols were completed with the consensus of the members of the SETI Committee in 1989. The document was approved by both the IAA Board of Trustees and the Board of Directors of the International Institute of Space Law. Over the next few years, the Post-Detection Protocols received the endorsement of several international organisations, including the International Astronautical Federation, COSPAR, the International Astronomical Union, and Commission J of the Union Radio Scientifique Internationale. In 2000, Jill Tarter and Ernst Fasani presented the Reply Protocols to the UN COPUOS. The Report of COPUOS noted that on the basis of the presentation, the Committee agreed that a copy of the position paper should be retained by the Office for Outer Space Affairs (UN Doc. A/55/20, para. 16) and that the Committee and its Legal Subcommittee should give serious consideration to the issue of an international process relating to possible communication with a discovered extraterrestrial civilisation (para. 157). The Report of

COPUOS was endorsed by the General Assembly (UNGA Res. A/RES/55/122 (27 Feb 2001)).

The widespread acceptance of the Protocols by these international organisations is indicative of the broad representation within the IAA, and the SETI Committee in particular, of individuals and representatives of institutions from the global SETI community. It also is illustrative of the way the Post-Detection Protocols fulfil one of the goals sought by the SETI Committee, namely the voluntary acceptance by the international community, whether any institution or individual participated in the preparation of the document.

The Post-Detection Protocols primarily were directed toward the verification of a potential candidate radio signal, and in the event of confirmation the privilege of making the first public announcement is reserved to the discoverer. It also provided that international consultations should be conducted prior to responding to a confirmed detection of ETI. The draft Reply Protocols picked up where the Post-Detection Protocols left off and set out aspirational guidelines for such international consultations to be conducted within the framework of the UN Committee on the Peaceful Uses of Outer Space and other governmental and non-governmental organisations. In addition, the Reply Protocols provided that the decision whether to reply, and if so, the content thereof, should be determined by the General Assembly. States are encouraged to not send a reply on a unilateral basis, nor to cooperate with efforts to communicate with ETI not in conformity with the Reply Protocols.

Although the Reply Protocols were approved by the SETI Committee in 1995, the substance was recognised as controversial and lacking substantive consensus. The document was labelled as a “Draft” and was not intended to be the final statement on these matters; rather, further discussion and work were anticipated. The Post-Detection Protocols provided that in the event of the discovery of credible evidence of ETI, “an international committee of scientists and other experts should be established to serve as a focal point for continuing analysis of all observational evidence collected in the aftermath of the discovery, and also to provide advice on the release of information to the public.” It was recommended that the SETI Committee initiate and maintain a list of individuals and institutional representatives willing to serve in such capacity.

The SETI Committee did not directly implement this suggestion, as a list of willing participants was not instituted as set out in the Post-Detection Protocols. However, the SETI Committee created a Post-Detection Subcommittee, and the membership roster was a de facto list of willing persons, albeit far from comprehensive or complete. In 2008 the Post-Detection Subcommittee, co-chaired by Paul Davies (US) and

Carol Oliver (Australia), began a review of the Post-Detection Protocols, and a draft was presented that year to the SETI Committee at the 59th IAC in Glasgow. Work on the redraft of this document continued over the next two years, and the updated Declaration of Principles was approved by the SETI Committee during the 61st IAC in Prague in 2010. However, the 2010 Declaration of Principles Concerning the Conduct of the Search for Extraterrestrial Intelligence neither sought nor received widespread institutional or organisational approval.

The pace of technological developments for SETI has accelerated in the past decade with far-reaching ramifications for the Protocols, including but not limited to the discovery of thousands of exoplanets, the global expansion of entities and institutions engaged in the study of SETI, and the explosion in social media as a pervasive presence in society. It became apparent within a short time after adoption of the 2010 Declaration of Principles that the document itself would soon become dated as the technology continued to advance and new discoveries, theories and methods emerged. The revolution in communications, exemplified by social media, has the benefit of facilitating the flow and exchange of information, but also has the detriments of being capable of disseminating false information at lightning speed and being used as an instrument of harassment against individuals and institutions [8][9][10][11].

At the meeting in Paris in March 2014, the SETI Committee took up a review of the Protocols and noted that the primary controversy at that time related to messaging to an ETI (METI). The PC approved the creation of a working group for revisions to the Protocols, referred to as the Subcommittee on SETI and METI. It was to be chaired by Alan Penny, with Guillermo Lemarchand and Daniela de Paulis as co-chairs [12]. At the next meeting of the PC in October 2014 during the 65th IAC in Toronto, the PC unanimously agreed to establish a new Protocols Subcommittee to “subsume the Post-Detection Group and METI protocol issues” [13].

These initiatives did not produce the desired result and the SETI Committee took up the matter again in 2017, appointing Duncan Forgan and Leslie Tennen to lead a renewed initiative. An ambitious and detailed strategy was developed which envisioned a dedicated website as a platform for interested parties to exchange ideas and participate in the revision process [14]. This initiative required institutional support which, unfortunately, could not be secured. However, Kathryn Denning, Anamaria Berea, Klara Anna Capova, Eric Korpela, Michael P. Oman-Reagan, and Jill Tarter joined Tennen and Forgan in preparing a comprehensive research roadmap for post-detection matters, which was presented to the 70th IAC in Washington, D.C. in 2019.

[15]. A few years later, the SETI Committee Executive went through a renewal process with a new chair, Michael Garrett, and two new co-chairs, Andrew Siemion, and Carol Oliver, with the updating of the SETI Protocols being high on the agenda, as mentioned in the introduction.

Towards the end of 2022, the SETI Committee Executive led by Michael Garrett (UK), recognising the 2010 version of the Protocols was badly in need of an update, established a new Task Group chaired by Kathryn Denning (Canada) and Carol Oliver (Australia, and one of two co-chairs of the SETI Committee). In the months following initial discussions between Denning and Oliver, Leslie Tennen, Garrett, and Andrew Siemion (US, also co-chair of the SETI Committee) were enlisted as the founding members. The remainder of this paper describes the work so far on an update. We outline other perspectives drawing on the work prior to the initiation of the new Task Group, and in several contexts of the reality that techno-signature and biosignature research are continually evolving. Among the contexts of updating the protocols is increasing international interest in preparing for the discovery of extraterrestrial life. This includes: the establishment of the SETI Post-Detection Hub at the University of St Andrews [16][17], an International Astronomical Union/ Kavli Foundation symposium to be held at the University of Durham in April 2024, "(Toward) Discovery of Life Beyond the Earth" [18]) with a significant component focused on post-detection; Penn State SETI Symposia with focused post-detection breakout sessions in 2022 and 2023 [19][20]; NASA Astrobiology workshops concerning standards of evidence for biosignatures [21][22], and the communication of biosignatures research results [23]. Beyond the scope of the present paper, but always in the background as Protocols revisions and post-detection challenges are considered, are societal dimensions of the search for life beyond Earth [24][25][26], and considerations from science communication and risk communication.

3. Scope of the Review of the Protocols

As noted above, the Protocols were originally prepared with specific reference to the search for radio signals, which was the predominant medium for conducting SETI at the time. Today the realms of search have expanded to encompass techno-signatures, including biosignatures, and while that may seem to be all-inclusive nomenclature, the Protocols should be expansive and not limited by restrictive definitions of contexts for searches or forms of detection. A revision to the Protocols should consider and seek to reflect general agreement on the criteria for basic matters such as what will constitute confirmation of discovery of ETI.

The Protocols are intended to provide guidance in the conduct of individuals and institutions engaged in SETI activities but are not a contract nor binding agreement. The effectiveness of the Protocols is dependent on the voluntary adherence by the global SETI community, which has as its foundation an element of good faith. Even if there was an effective means for ensuring compliance, it is entirely possible, even probable, that some individuals would fail to act in conformity with the accepted norms and processes when faced with real-world pressures and motivations. No mechanism can truly prevent ill-timed leaks concerning work in progress, or premature announcements, or misinformation. However, the IAA, through the SETI Committee, must serve as a credible and reliable authority to offer broad guidance based on its long history in considering post-detection and the expertise it has built and as searches for techno-signatures proceed in the context of significant ongoing public interest.

As noted, the Post-Detection Protocols and the 2010 Declaration of Principles refer to "parties" or "signatories." The Declaration of Principles provides that a list of signatories will be maintained by the IAA. Both documents were adopted by the SETI Committee, and while the Post-Detection Protocols have been approved or endorsed by several international organisations, neither document was opened for signature in the traditional sense, and thus neither document has any formal "parties" or "signatories." The adoption of the Protocols by the SETI Committee was a de facto, but ersatz, acceptance of the instruments by the individual members of the committee and the institutions they represented. The widespread acceptance of the Post-Detection Protocols attested to the importance the endorsing organisations placed on the document, adding weight as a guiding document for individuals and institutions in considering its terms. The review and redraft of the Protocols should continue this practice and consider procedures for formal acceptance and endorsement by individuals, institutions, and organisations, and for the reporting of the current lists on the IAA website. Some have called for the drafting of a treaty on these matters. While the possibility of the adoption of a treaty on this (or any other space related matter) can be debated, a comprehensive revision to the Protocols can serve as a form of soft law pending the adoption of a formal inter-governmental agreement.

The purpose of the Protocols is to articulate a set of aspirational principles to guide the intentions and activities of individuals and institutions within the global SETI community. This includes standards of conduct to promote transparency, dissemination of data, scepticism, multiple verifications, and other activities of a scientific nature, as well as responsible communication with the world. This latter aspect was

not adequately considered or reflected in the Post-Detection Protocols or the Reply Protocols, but the radical transformation in the dissemination of information ushered in by the Internet has raised the level of concerns relating to interactions with the media such that it is appropriate for the Protocols to include social media safety guidelines for researchers.

The Protocols can articulate the fundamental principles as baselines of behaviour, and supplemental guidance can be provided in the forms of Codes of Conduct and Best Practices guidelines. The development of these codes will rely on qualified research teams with expertise in areas such as social media, public understanding of science, emergency communications, and graphics. These documents will be subject to continuing review and revision to adapt to rapidly changing circumstances.

The 2019 Roadmap [27] identified certain factors as basic to the post-detection terrain that the Protocols must now acknowledge, including these:

- The contexts of techno-signatures and biosignatures research are continually evolving.
- The potential forms of detection are continually evolving and expanding.
- There is incomplete agreement among scientists on what would constitute confirmation of a discovery of ETI.
- The process of confirmation could be prolonged (and the world doesn't wait).
- There is no mechanism by which to prevent people making premature announcements.
- We cannot prevent misinformation being spread.
- We cannot fully protect scientists, citizen scientists, or students, from online or real-life abuse such as doxxing, swatting, or worse.

4. The Changing Science of Techno-signatures: Expansion, Internationalisation, Intensification, and New Possibilities

SETI has undergone a remarkable transformation over the last decade. The establishment of the \$100M Breakthrough Listen programme in 2015 has been one of the driving forces that has rejuvenated the field, also in the wider global context [28], alongside the long-established SETI Institute, which is a leading light in community engagement and education in addition to its search program on the Allen Telescope Array in California. The size of the research community has grown significantly, attracting a new generation of talented astronomers and engineers who are utilising a wide range of new and different instruments and techniques. The sensitivity of searches has greatly increased, but there has also been a diversification of

potential techno-signature discovery types, which has direct relevance for the Protocols.

Commensal SETI surveys using radio interferometer arrays such as the Very Large Array (VLA) and the Karoo Array Telescope (MeerKAT) are preparing the way towards exploitation of the next generation of radio telescopes, the Square Kilometre Array (SKA) being one example. The raw SETI search capability has also increased by many orders of magnitude, with state-of-the-art backends connected to the largest radio telescopes in the world, including the Green Bank Telescope in the USA and the 64-m Parkes Telescope in Australia. SETI programmes are also being developed at many other radio observatories around the world, including the Allen Telescope Array in the US, the Murchison Widefield Array (MWA) in Australia, the FAST in China, the Giant Metrewave Radio Telescope (uGMRT) in India, the enhanced Multi Element Remotely Link Interferometer Network (e-MERLIN/EVN) in the UK, the Low Frequency Array (LOFAR) Telescope (especially the Swedish and Irish stations), Nancay in France, and the Sardinian and Medicina telescopes in Italy, to name just a few. Advances in searches for narrow-band optical laser transmission have opened new areas of parameter space, with surveys covering large areas of sky with very short (sub-second) exposure times employed.

Numerous technical advances have benefited significantly from recent breakthroughs in digital and computer processing, primarily driven by commercial applications. The field generally has also profited from a more methodical approach as adopted by large professional astronomical all-sky surveys – a greater emphasis on uniform post-processing methods, careful data analysis supported by AI, and the publication of results in high impact, peer-reviewed journals is evident. In this way, SETI researchers are very much embedded in the wider astronomical community, and the intersections between fields are growing – examples include Exoplanets, Fast Radio Bursts, Multi-messenger, and transient phenomena in general. This interconnectivity has significant implications for the Protocols.

The traditional search for powerful narrow-band extraterrestrial radio beacons has been augmented by a much more extensive approach that looks for transmissions not intentionally directed at us, with a more generalised goal of identifying so-called “techno-signatures” – anomalies in astronomical data that don't have an obvious natural, astrophysical explanation. Techno-signatures are the imprint of extraterrestrial technology on astronomical data, comparable to the biosignatures that are indicators of the presence of biological life in the spectra of exoplanet atmospheres. As a result, the search itself has expanded beyond the radio frequencies of the spectrum, into the infrared and

optical domains. A consequence of this is that not all techno-signatures will be immediately recognised as such – while the detection of an extraterrestrial narrow-band radio signal is a relatively clear-cut case, it would be harder to prove a claim that unexpected infrared excess in a nearby star or exoplanet is due to the waste heat generated by an advanced technical civilisation. The latter claim could take considerable persuasive data to convince others of its validity. Some potential techno-signature discoveries may therefore be subject to a prolonged timeline of years or even decades for confirmation or disconfirmation, which would pose particular challenges for public communication about scientific debates or non-consensus situations, amid ongoing research.

Another major development is that many of the next generation of telescopes are really survey instruments, able to observe large fractions of the sky on timescales of a few days (e.g., LSST, SKA, EUCLID). The data deluge being generated is something that astronomers have been grappling with over the last few years, but more challenges are just around the corner. Increases in data volume via greater sky coverage are now also accompanied by greater data complexity – survey instruments offer combinations of good temporal and spectral resolution, high cadence, and full polarisation information. Doing justice to all the data generated will take a major effort from the astronomical community, especially in terms of follow-up measurements. It is predictable that the kind of data being delivered will generate many millions of anomalies every day, and some of these will be relevant to SETI. What is already clear, is that astronomers working with survey instruments will have a role in the SETI pursuit whether they intend to or not! This explosion in survey science, and the wide variety of techno-signatures that extraterrestrial societies might generate, suggests that a valid techno-signature is more likely to be discovered by an astronomer minding their own business, who stumbles over something weird in the data. At some level this has always been the case, but the odds are now stacked in favour of a detection being made by a group or even a large team with no or very limited experience of SETI or the SETI detection protocols. In addition, to note that throughout the history of the search for techno-signatures, there are no confirmed discoveries to date, but that these null results are useful in providing some constraints, for example, there seem to be no loudly transmitting neighbours.

5. Issues for revising the Protocols: Lessons from the recent past, and current perspectives from the SETI community

The ostensibly simple task of updating the 2010 Declaration of Principles is not straightforward. In addition to major developments in SETI/techno-

signatures search methods and technologies described above, and the current extremely dynamic human-to-human communication environment on Earth, we have many matters of definition and scope. Even updates to terminology instigate a cascade of issues: as just one example, “the search for techno-signatures” is now a standard term increasingly superseding “SETI” and should thus be included in an updated document, but in some usages, the breadth of the term “techno-signatures” includes areas which the astronomy-focused original Protocols were not designed to handle, and which the SETI Committee may not be ideally situated to address. Therefore, an update to the Principles requires some careful deliberation and community consultation, even if its objective is simply to preserve the original intentions. This element of deliberation/consultation has always been a part of Protocol development/revision. Similarly, the tripartite structure we envision, discussed above – i.e., the timeless Principles plus the more agile/responsive Codes of Conduct, and Best Practices – should make future updates easier, but initially developing this structure requires some conceptual work concerning the best place to address any issue.

As we proceed, we should draw upon the decades of experience since the Protocols were first envisioned and established. Not only do we have many prior examples of claimed detections of extraterrestrial life stretching back into the 1800s – see Denning et al 2019 and Dick 2018 [29][30] for discussions of what can be learned from these – but just since the Protocols were first developed, there have been multiple ‘false alarms’, i.e., premature leaks or announcements concerning potential SETI discoveries. Prominent examples have included YZ-Cet [31][32], EQ-Peg [33][34], Oumuamua, Boyajian’s Star, HD 164595/Ratan-600 [35], Borra and Trottier [36], FRBs, and BLC-1.

Analysis of such instances is assuredly incomplete, but one thing is clear: they have tended to demonstrate that desirable though it might be, the idealised sequence in the Protocols of ‘first consult within the scientific community, verify, and *then* tell the world’ will not generally work in practice. Intense interest from journalists and activity on social media rapidly follow any hint or inkling of a potential discovery, and rumours must be addressed to reduce the proliferation of unfounded speculation and to provide sound grounding in the best science available. A vivid example was provided recently by the candidate signal BLC-1, with its extensive media coverage [37][38][39][40][41][42].

Of note, the 2023 art project, A Sign in Space, simulating an incoming extraterrestrial transmission, also generated extraordinary global public interest and participation, from which much can be learned [43].

Thus, in these multiple ‘dry runs’, we have seen that scientific collaboration, follow-up studies on candidate

signals or detections, and communication about these processes with media/members of the public, unfold *concurrently* to some degree. And in some future putative detection situations, consensus may be elusive, and verification processes may be prolonged. The scientific community's broader responsibilities are therefore more complicated than the original Protocols envisaged – and discharging those responsibilities is particularly challenging in a communications environment dominated by social media, with its well-known incivilities, risks, and tendencies to amplify dissension and spread misinformation. In terms of strategies for disseminating accurate and reliable information and countering misinformation, there is much to be learned from other examples of high-impact science communication in recent years, for example, in the cases of climate science and the COVID-19 pandemic.

On the other hand, the many high-profile premature claims of the discovery of extraterrestrial life in recent decades – the examples above, but also some from the world of astrobiology, from ALH84001 to GFAJ-1 to Venusian phosphine [44][45][46] – have contributed to our current interesting situation with public perceptions about ETL, that is, the widely held popular perception that life beyond Earth definitely exists and/or has already been discovered, demonstrated in a global survey report by Lampert and Papadongonas [47]. Such popular beliefs could dampen some of the societal impact of any eventual confirmed findings, but nonetheless we should not underestimate the potential impacts from some types of techno-signature discoveries: the details of a discovery would certainly matter, and so would the presentation of those details.

Recent years have also seen the production of abundant scholarship and commentary about potential post-discovery issues, from a multitude of disciplinary perspectives. This body of work is too extensive to fully represent here, but multiple authors have carefully identified limitations in the Protocols, and/or have recommended additional risk analysis, regulations, policies, treaties, public consultations, etc., to cover potential ramifications of a discovery ranging from the political to the practical [48][49][50][51][52][53][54][55][56][57][58]. Works such as these suggest areas which may ultimately be adjacent to future Protocols or may have some articulation with them, or incorporation within them; these should be consulted in detail as our deliberations about revisions proceed.

With respect to Protocol updates, we note that in comparison to 15 years ago, when the last revisions were undertaken, today the SETI Committee and the SETI community are *considerably* larger – but also better connected online through long-distance collaborations and virtual meetings, particularly since the COVID-19 pandemic-driven transition to remote

work. It will thus be possible to have international conversations more effectively, not restricting crucial conversations to email exchanges or in-person meetings which not everyone can attend. At the same time, there are limitations: limited diversity is an issue within the SETI Committee composition, and, of course, we cannot regard the SETI Committee as a body that represents the interests of all stakeholders worldwide.

Mapping out the areas of concern for SETI community members is a necessary initial step in a Protocols update. Therefore, informal conversations with SETI practitioners and SETI researchers were initiated in 2023 by Denning and other members of the Task Group, as a starting point. We sought input about which aspects of the existing Protocols should be preserved, which should be modified, and which SETI-related matters should potentially be addressed by agreements or instruments other than the Protocols. Additional structured community consultations will follow. These preliminary inquiries have yielded the following points (note this list is not exhaustive, and order of items does not reflect their relative importance):

Consensus and division: There is strong support for the existing ideals of: careful scientific research and evaluation of evidence; transparency; responsible communication with the media, public and governments; and restraint from unilateral actions (e.g., transmissions) in the event of a confirmed detection. However, strong, opposing, and difficult-to-reconcile views are held on some issues such as METI (de novo transmissions), and so achieving community consensus on those matters will be challenging.

Terminology: recommendations include the replacement of “extraterrestrial intelligence” with “extraterrestrial technology”, and the use of the term “techno-signatures” instead of “SETI”.

Signatories for Protocols: Some would like to see the opportunity and process for individuals and institutions to publicly identify themselves as signatories. Others ask whether we could incentivise private actors (individuals and organisations) to take this pledge.

Astronomical techno-signatures other than radio: Many advocate for the inclusion of optical and archival SETI, and exoplanet-related techno-signatures.

Techno-signatures within our solar system: There is concern about the potential for discovery of solar system techno-signatures (e.g., objects in space or on moons, planets, or asteroids), which would not be sufficiently covered by existing Protocols, and should be addressed.

Techno-signatures within Earth's atmosphere or cis-lunar space: Ongoing and increasing interest in UAPs, particularly but not only in the USA, prompts suggestions about the inclusion or exclusion of this area

from our remit: some suggest that the region from Earth's atmosphere up to the lowest cis-lunar orbits should be excluded from the Protocols, because so much human technology exists in this zone, some of it secret; others suggest defining the kind of UAP science result, which would constitute a detection of extraterrestrial technology; still others note that the UAP issues are so complex that they warrant an entirely separate and dedicated framework.

Protection of the mode/mechanism of discovery: The Post-Detection Protocols provided in paragraph 7 for the protection of pertinent radio frequencies in the event of a detection of an electromagnetic signal. Consideration should be given to guidelines for the protection of other data sources, as applicable. It must be recognised that the private sector will be conducting activities on the Moon and other celestial bodies and could inadvertently discover evidence of alien life or technology, in which event specific obligations must be developed for the protection of artefacts and the vicinity of the discovery to prevent disturbing, altering, or damaging the site pending scientific investigation [58][59].

Transmissions from Earth, de novo and reply: METI continues to be a highly contentious issue, as it has been for years [60][61][62][63][64][65][66][67][68][69]. It is noteworthy that, subsequent to the last Protocol revision, some SETI scientists chose in 2015 to issue their own public statement regarding METI, addressing active SETI/de novo transmissions as well as reply transmissions: "Intentionally signalling other civilisations in the Milky Way Galaxy raises concerns from all the people of Earth... A worldwide scientific, political, and humanitarian discussion must occur before any message is sent." [69]. Today, some are strongly opposed to METI, some are strongly in favour, some emphasise specific conditions, and some are focused upon principles and others are focused upon pragmatic aspects of risk. Some still prefer the 1989 Principles language to the 2010 revision. Some specify concern about the situations in which transmissions from Earth could have a practical near-term impact (for example, attempts to communicate with a probe in our solar system). In terms of replies, some suggest that we should do more than vaguely indicate the need for a vote: e.g., we should define the questions that need to be answered regarding sending a reply (for example, how close is the sender, is the transmission clearly an attempt to communicate, is it likely that the sender is aware of our existence, etc.) Others observe that considerable thought should be given to the matter of who will be asked for guidance or consent, and equity in this regard. Others note that given the ease of transmission now, we should assume that some non-signatories, including members of the public, will proceed to transmit in the direction of an incoming signal if they have the location

information and the capability to transmit, and therefore suggest we should consider protecting some location information.

Communication with media, public, and other stakeholders: There are substantial concerns about social media, including sentiments that scientists should not have to engage in this medium if they prefer not to. Some note that research organisations should shield researchers from excessive online exposure and be proactive about researcher safety. Some emphasise the importance of managing media and public expectations of scientists when essential, time-consuming verification work is underway (that is, allow scientists to simply say that they don't have an answer yet and need to continue their work). Some suggest that SETI organisations should jointly operate a system such as the Interstellar Communication Relay [70], which could serve as a reliable news distribution site in pre-detection mode, and on very short notice can serve as a reliable, scalable, and resilient source of information concerning significant discoveries. Others also encourage the development of a trustworthy expert voice, and steps to make it influential. Others underscore that media and public responses will not all be positive/favourable. Further recommendations include 'damage-control' strategies or backup protocols in case the intended good practice is not followed by all. Some express an interest in better integration of science communication expertise into our work, to help optimise communication about candidate detections, particularly about uncertainties which may be either short-lived or unresolvable, about situations where there is no strong consensus, and in general about the extended nature of scientific discovery. Several commented on the Rio Scale as insufficient, despite its inclusion of some useful principles concerning types of discovery and confidence in evidence. Others suggest that pre-registering research hypotheses might be a useful form of transparency which would help pre-empt leaks and ensure credit to discoverers.

Communications security: Some suggest that SETI organisations and principal researchers should secure their communication and social media services routinely, and particularly during a confirmation process, to minimise potential damage from hackers: for example, enable multi-factor authentication, protect web services against Distributed Denial of Service (DDOS) attacks, etc.

Protection for observatories: Upon a confirmation of detection, SETI-capable observatories could be the target of interference or sabotage, and it was suggested that protection should be requested from the UN or host nations.

Reporting mechanisms to protect discovery credit are needed. Various mechanisms for this have been suggested.

Enforceable policies: Bearing in mind the inherent limitations on enforceability at both the international and national levels, the articulation of carefully considered and foresighted standards can serve to also guide national authorities vis-a-vis the activities of students, schools, citizen scientists, unaffiliated individuals, and other private actors [71].

Standards of evidence, verification, and consensus: Scientific discovery is increasingly recognised to be an extended process, not a moment [72], and this process may be prolonged for some potential detections of extraterrestrial life or technology. Some emphasise that consensus may be elusive or long-delayed in some potential techno-signature discoveries, and that there could be disputes within the techno-signatures community, or with other scientists who have differing/competing priorities. Some respondents aspire to unification of biosignatures and techno-signatures frameworks for verification and communication of levels of confidence in detection. Some emphasise the need for techno-signatures scientists to continue work on establishing standards of evidence and confirmation procedures, and to continue work on search-specific verification frameworks [73][74].

Post-detection Task Group within the IAA SETI Committee: This does not currently exist, though explicitly mentioned in the 2010 Declaration of Principles: some respondents strongly encourage its reconstitution, and/or the support of groups with a similar focus, given the essential nature of post-discovery-focused research, policies, and interdisciplinary expertise. (We note that Goodman's original proposal for post-detection protocols and their implementation explicitly suggested an interdisciplinary Standing Consultative Committee with five-year terms with financial and administrative support to enable their participation, as an integral part of the proposal [75]).

Other post-discovery considerations: A science-policy interface is necessary. Reporting to the S-G of the UN is laudable, but what should the UN then do? The IAA has COPUOS observer status, but there is no fast-response unit. There is currently no science advice mechanism for the UN (though one may be in development). Is it enough to simply issue a coherent initial announcement, or are there further responsibilities or roles for scientists?

Data management: Data storage for the short and long term (protection against loss, hacking, corruption) is a major concern, and there needs to be a serious discussion about overall data strategy (collections, replications, analyses, maintenance, digitalisation, safety, and archiving). Financial or other forms of opportunistic gain might play a role in the use of data of an extraterrestrial signal; manipulation or corruption are possible outcomes. Who owns the data, and who is responsible for managing the data and preserving their

integrity? There are also questions about exactly how much data should be available to the public, and how the provision of data to the scientific community would work in practice.

Degrees of transparency: there may be differences of opinion about the implementation of the principle of transparency in different contexts, for example, during the verification process versus after a confirmation of a discovery. There is also a concern that incomplete transparency may contribute to public mistrust and/or backlash, while others note that some data (for example, precise locations/frequencies pertinent to discoveries) should potentially be restricted to scientists to prevent a cavalcade of unilateral or unsanctioned actions.

Who will be involved in a discovery: Multiple respondents emphasise the possibility that someone who is not actually conducting a SETI search may find evidence of extraterrestrial technology, and the possibility that these individuals may not know anything about the Protocols. Some suggest that we should proactively address the problem of actors undertaking activities with the potential for serendipitous discovery.

New Ethical Problems: In contrast to the earliest SETI searches, which were essentially only capable of discovering targeted transmissions specifically intended for Earth, new techno-signatures searches and future astronomy could detect a wide range of techno-signatures without implicit or explicit evidence concerning the intentions of the entities or society responsible for those phenomena. We may not know from initial data, and perhaps not from follow-up studies either, whether they intended to attract our notice or not, would welcome or prefer to avoid contact with us, know of our existence, coexist with us in time, etc. This requires ethical analysis and further consideration as we formulate updated protocols.

Clearly, SETI community members have given such matters considerable thought – we have only scratched the surface here – and we look forward to further discussions. Part of our future work will be decisions about the following: which issues should be incorporated soon in a revised IAA SETI Declaration of Principles (either generally acknowledged in a Preamble, or specifically addressed within a list of Principles); which issues might instead be addressed in an additional Declaration or Statement at some point; which issues should be addressed with Codes of Conduct and Best Practices documents; and which issues should be explicitly left aside for other groups to tackle. Below, Appendix A, Table 1: Example of a Potential Structure for Protocol Updates is meant to *illustrate* such a process of disentanglement, *not* to be definitive in its content.

6. Conclusions

Our ultimate aim in this process is to: produce a distilled set of Principles which cover essential tenets of technosignatures searches and will not require frequent updating; identify areas which can be addressed through more regularly updated Codes of Conduct and/or Best Practices documents which are responsive to the rapidly changing communications landscape and scientific research environments; and carefully delineate areas related to extraterrestrial life which the IAA SETI Committee will *not* address explicitly in its Protocols, and which may be better handled by other groups or committees.

As we work towards a revised set of Principles, we anticipate using a modified Delphi process for

structured consultation with SETI practitioners and researchers in related fields, to gather expert viewpoints and carefully identify specific areas of concern. We anticipate workshopping those specific problems as necessary, and an iterative process with opportunities for input on draft language. We hope that such a process will lead to the broadest acceptance possible in our future Protocols update.

Acknowledgements

The authors thank their colleagues in the SETI community who have provided their input in our informal discussions thus far, and those who worked on previous versions of the Protocols.

Appendix A

Table 1: Example of a Potential Structure for Protocol Updates

**See the original Protocols at reference [3, 4, 5]*

Essential updates to include soon in a new Preamble to the Declaration of Principles Concerning the Conduct of the Search for Extraterrestrial Intelligence	* Contextualise the Protocols, i.e., point out their specific goals and the underlying reasons for their form/content, clarify whom they are addressing, and note the possibility that a discovery may come from outside the scientific community * Clear definition of areas addressed and <u>not</u> addressed by these Protocols * In the first line, specify a little more about the “scientific Search for Extraterrestrial Intelligence”, what it does and doesn’t involve * Reference to external efforts to study social impacts etc., and support of these in principle * Reference to external efforts to improve science communication in relation to ETL/ETI, and support of these in principle * Reference to scientists’ efforts re: technical aspects of verification frameworks etc. * Recognition that the scientific and communication landscapes are dynamic, and that these Principles require augmentation with updateable Codes of Conduct and Best Practices. Recognition that the production/maintenance of such guidelines requires a sustained community effort and interdisciplinary expertise. * Give the opportunity for signatories, individual and institutional, to declare themselves * Include reference to a living bibliography or ‘supporting documents’ to build on history and not reinvent the wheel * In general: if these Principles are intended to be “user-friendly” to a wider community than the original collection of signatories / SETI researchers, then more explanation is needed for clarity
Essential updates to the 2010 Declaration of Principles Concerning the Conduct of the Search for Extraterrestrial Intelligence	* Terminology updates (e.g., ‘techno-signatures’) * Point 1, better define transparent * Point 2, concerning candidate evidence: should incorporate the recognition that consensus may be elusive, and a study process may be prolonged * Point 2: concerning candidate evidence: should incorporate the recognition that communication may be necessary about verification efforts underway to dispel rumours, but manage expectations (i.e., scientists are not endlessly available, they have their work to do) * Point 2: The Rio Scale requires augmentation for optimal science and risk communication, i.e., is not in itself sufficient. Could replace with language such as “best practices from science communication available should be used as a guide”, linking to Best Practices guidelines updated

	annually. * Point 5: Some elaboration about safe/reliable/resilient data practices may be warranted; perhaps also elaboration about whether access for “further analysis and interpretation” will be restricted to the scientific community (how is that defined?) or broader * Point 6: Consider extension of the principle of source data protection for discoveries not based in radio frequencies * Point 7: a Post-Detection Task Group currently does not exist within the IAA SETI Committee, so this needs to be reconciled. * Point 8: refraining solely from unilaterally responding to signals is not sufficiently inclusive given the range of potential detections and subsequent actions (e.g., recovery of artifacts)
Longer-term discussions within IAA SETI PC / community	* METI: De novo, and Reply protocols * Some solar system technology discoveries (though some may prove too complex for IAA SETI alone)
Codes of Conduct	*Codes of Conduct from international science organizations might be usefully supplemented with codes specific to techno-signatures-related research, e.g. specifying the need for exceptional care with verification procedures and public communication.
Best Practices	For example, * Science and risk communication (strategies in addition to the Rio Scale) * Interstellar Communication Relay * Data storage * Verification frameworks (will be specific to different search types) * Recording SETI searches in central repository * Nomenclature updates
Leave for other processes / groups to tackle	* UAPs * Some solar system technology discoveries (some may prove too complex for IAA SETI alone) * Protocols for managing actual contact, should it occur * Some aspects of METI

References

[1] L. Walton, The History of the IAA SETI Permanent Committee – 1980 to 1989, IAC-19, A4.2.5, x52248 (2019).

[2] C. Maccone, SETI and the IAA SETI Permanent Committee: Past, Present and Possible Future, in Private Law, Public Law, Metalaw and Public Policy in Space, A Liber Amicorum in Honor of Ernst Fasan, p. 145 (P.M. Sterns and L.I. Tennen eds. 2016), Springer, Cham.

[3] Declaration of Principles Concerning Activities Following the Detection of ExtraTerrestrial (sic) Intelligence, <https://iaaspace.org/wp-content/uploads/iaa/Scientific%20Activity/setideclaration.pdf>.

[4] Draft Declaration of Principles Concerning Sending Communications with Extraterrestrial Intelligence, <https://iaaspace.org/wp-content/uploads/iaa/Scientific%20Activity/setidraft.pdf>.

[5] Declaration of Principles Concerning the Conduct of the Search for Extraterrestrial Intelligence, <https://iaaspace.org/wp-content/uploads/iaa/Scientific%20Activity/setideclaration2.pdf>.

[6] A.E. Goodman, Diplomatic and Political Problems Affecting the Formulation and Implementation of an International Protocol for Activities Following The Detection of a Signal from Extraterrestrial Intelligence, Acta Astronautica 21(2) (1990) 103-108.

[7] J. Tarter, M. Michaud, Preface to Special Issue of Acta Astronautica 21(2) (1990) xi-xii.

[8] P.M. Sterns, L.I. Tennen, SETI, Metalaw, and Social Media, in: Private Law, Public Law, Metalaw and Public Policy in Space, Springer, Cham, 2016a, 159–179

[9] P.M. Sterns, L.I. Tennen, Social Media And The SETI Post-Detection Declaration Of Principles, IAC-16- A4.2.7.33638, 2016b

- [10] D. Forgan, A. Scholz, #Found Them – 21st Century pre-search and post-detection SETI protocols, *Acta Astronautica*, 126, September-October 2016.
- [11] C. Oliver, Social Networking: Implications for Post-Detection Communications, IAC-10, A4.2.5, x7167, 61st IAC, Prague, 2010.
- [12] Minutes of the SETI Committee, March 20, 2014, <https://iaaspace.org/wp-content/uploads/iaa/Scientific%20Activity/setiminutes14a.pdf>
- [13] Minutes of the SETI Permanent Committee <https://iaaspace.org/wp-content/uploads/iaa/Scientific%20Activity/setiminutes14a.pdf>
- [14] L.Tennen, D. Forgan, Review of the SETI Post-detection and Reply Protocols: Current Actions And Developments, IAC-18,A4,2,1,x47585 69th IAC, Bremen, Germany, 2018, 1-5 October.
- [15] K. Denning, A. Berea, K.A. Capova, D. Forgan, E. Korpela, M.P. Oman-Reagan, J. Tarter, L.I. Tennen, SETI and Post-Detection: Towards a New Research Roadmap. IAC-19-A4.2.4.x52210, 70th IAC, Washington DC, 2019.
- [16] <https://seti.wp.st-andrews.ac.uk>, 2022 news release, www.eurekalert.org/news-releases/970047
- [17] J. Elliott et al., The SETI Post-Detection Hub, International Astronautical Congress, Baku, 2023.
- [18] International Astronomical Union/ Kavli Foundation symposium, (Toward) Discovery of Life Beyond Earth, University of Durham (2024) 15-19 April. <https://kavli-iau-2024.durham.ac.uk/>
- [19] Kathryn Denning, B. McConnell, C. Haramia, G. Profitiliotis, R. Charbonneau, J. Tarter, S. Wright 2022, Post-detection for the 2020s: fundamentals, current concerns, and new discussions. Breakout session, Penn State SETI Symposium, June 2022. <https://sites.psu.edu/setisymposium2022/files/2022/06/Post-Detection-breakout-final-program-19-June-2022.pdf>
- [20] Kathryn Denning, Anamaria Berea, Rebecca Charbonneau, Steven J Dick, Chelsea Haramia, Brian S McConnell, & George Profitiliotis 2023, The Alien in the Machine: Post-Detection in the Era of AI. Breakout session, Penn State SETI Symposium, June 2023. <https://sites.psu.edu/setisymposium2023/agenda/breakout-sessions/>
- [21] V. Meadows, H. Graham, et al., Community Report from the Biosignatures Standards of Evidence Workshop.arXiv:2210.14293v2 [astro-ph.IM], <https://doi.org/10.48550/arXiv.2210.14293>, 2021.
- [22] Committee on Planetary Sciences, National Academies of Sciences, Engineering, and Medicine 2022. Independent Review of the Community Report from the Biosignature Standards of Evidence Workshop: Report Series—Committee on Astrobiology and Planetary Sciences. Washington, DC: The National Academies Press. <https://doi.org/10.17226/26621>.
- [23] Communicating Discoveries in the Search for Life in the Universe, upcoming February 2024, <https://astrobiology.nasa.gov/events/communicating-discoveries-in-the-search-for-life-i/>.
- [24] S. J. Dick, *Astrobiology, discovery, and societal impact*, Cambridge University Press, 2018.
- [25] K. Denning and S. J. Dick, Preparing for the Discovery of Life Beyond Earth, Astro2020 National Academies Decadal Survey white paper, 2019. <https://tinyurl.com/y64wqpqe>
- [26] J. Billingham, SETI Institute, Social implications of the detection of an extraterrestrial civilisation: a report of the workshops on the cultural aspects of SETI, SETI Press, Mountain View, Calif., 1999.
- [27] K. Denning, A. Berea, K.A. Capova, D. Forgan, E. Korpela, M.P. Oman-Reagan, J. Tarter, L.I. Tennen, SETI and Post-Detection: Towards a New Research Roadmap. IAC-19-A4.2.4.x52210, 2019.
- [28] S.P. Worden, J. Drew, A. Siemion, D. Werthimer, D. DeBoer, S. Croft, D. MacMahon, M. Lebofsky, H. Isaacson, J.Hickish, D. Price, V. Gajjar, J. T. Wright, Breakthrough Listen – A new search for life in the universe. *Acta Astronautica*, Volume 139, Pages 98-101, ISSN 0094-5765, 2017. <https://doi.org/10.1016/j.actaastro.2017.06.008>.
- [29] K. Denning, A. Berea, K.A. Capova, D. Forgan, E. Korpela, M.P. Oman-Reagan, J. Tarter, L.I. Tennen, SETI and Post-Detection: Towards a New Research Roadmap. IAC-19-A4.2.4.x52210, 2019.
- [30] S. J. Dick, *Astrobiology, discovery, and societal impact*, Cambridge University Press, 2018.
- [31] S. Scoles, *Making Contact: Jill Tarter and the Search for Extraterrestrial Intelligence*, Pegasus Books, New York, 2017, 200-202.
- [32] S. Shostak, *Confessions of an Alien Hunter*, National Geographic, Washington, 2009, 12-18.
- [33] C. Oliver, H. Sim, S.Shostak, The Case of EQ Peg: Challenge and Response, 50th International Astronautical Congress, Amsterdam, 1999, 4-8 October.
- [34] S. Shostak, *Confessions of an Alien Hunter*, National Geographic, Washington, 2009, 12-18.
- [35] Breakthrough Listen follow-up of a Transient Signal from the RATAN-600 Telescope in the Direction of HD164595 <https://seti.berkeley.edu/HD164595.pdf>

- [36] E.F. Borra, L. Trottier, Discovery of a peculiar spectra modulations in a small fraction of solar type stars, *Pub.ASP*, 128, 2016, 114201.
- [37] S. Smith, D. Price, S.Z. Sheikh et al., A radio technosignature search towards Proxima Centauri resulting in a signal of interest. *Nat Astron* 5, 1148–1152 (2021). <https://doi.org/10.1038/s41550-021-01479-w>
- [38] S.Z. Sheikh, S. Smith, D.C. Price, et al., Analysis of the Breakthrough Listen signal of interest blc1 with techno-signature verification framework, *Nature Astronomy*, 5 (11), 2021, 11-53-1162, arXiv:2111.06350. doi:10.1038/s41550-021-01508-8.
- [39] I. Sample, Scientists looking for aliens investigate radio beam 'from nearby star' 18 Dec 2020, www.theguardian.com/science/2020/dec/18/scientists-looking-for-aliens-investigate-radio-beam-from-nearby-star
- [40] J. O'Callaghan, L. Billings, "Alien Hunters Discover Mysterious Signal from Proxima Centauri" 18 Dec 2020, *Sci Am*, www.scientificamerican.com/article/alien-hunters-discover-mysterious-signal-from-proxima-centauri/
- [41] S. Shostak, A Signal from Proxima Centauri? 19 Dec 2020, www.seti.org/signal-proxima-centauri
- [42] Breakthrough Listen's First "Signal of Interest", <https://seti.berkeley.edu/blc1/>
- [43] D. De Paulis 2023 <https://assignin.space/>.
- [44] S. J. Dick, *Astrobiology, discovery, and societal impact*, Cambridge University Press 2018.
- [45] V. Meadows, H. Graham, et al., *Community Report from the Biosignatures Standards of Evidence Workshop*, 2021. Available: arXiv:2210.14293v2 [astro-ph.IM], <https://doi.org/10.48550/arXiv.2210.14293>
- [46] Committee on Planetary Sciences, National Academies of Sciences, Engineering, and Medicine 2022. *Independent Review of the Community Report from the Biosignature Standards of Evidence Workshop: Report Series—Committee on Astrobiology and Planetary Sciences*. Washington, DC: The National Academies Press. <https://doi.org/10.17226/26621>.
- [47] M. Lampert, P. Papadongonas, Majority of humanity say we are not alone in the universe: Values-based learning from the Glocalities survey in 24 countries, *Motivaction International B.V.*, 2017, Amsterdam. Available: <https://glocalities.com/reports/majority-of-humanity-say-we-are-not-alone-in-the-universe>.
- [48] K. W. Wisian, J. W. Traphagan, *The Search for Extraterrestrial Intelligence: A Realpolitik Consideration*, 2020, *Space Policy*, Volume 52, 101377, ISSN 0265-9646, <https://doi.org/10.1016/j.spacepol.2020.101377>.
- [49] J. Gertz, *Post-Detection SETI Protocols and METI: The Time Has Come To Regulate Them Both*, *JBIS*, 69, 2017, 263-270. Available at: <https://arxiv.org/abs/1701.08422>
- [50] J. Gertz, *Interstellar Diplomacy*. Forthcoming 2023 in *JBIS*. Available: <https://arxiv.org/abs/2308.14917>
- [51] S.J. Dick, ed., *The Impact of Discovering Life Beyond Earth*, Cambridge University Press, 2015.
- [52] S. J. Dick, *Astrobiology, discovery, and societal impact*, Cambridge University Press ,2018.
- [53] Brian McConnell (2020). *The interstellar communication relay*. *International Journal of Astrobiology* 19, 419–422. <https://doi.org/10.1017/S1473550420000178>
- [54] P. Hatfield, L. Trueblood, SETI and democracy, *Acta Astronautica*, 202, 180, 2021, 596-603, ISSN 0094-5765, <https://doi.org/10.1016/j.actaastro.2020.11.053>.
- [55] M. Bohlander, *Contact with Extraterrestrial Intelligence and Human Law: The Applicability of Rules of War and Human Rights*. Brill Nijhoff, 2023.
- [56] A. Anton, M. Schetsche, *Meeting the Alien: An Introduction to Exosociology*, Springer, 2023.
- [57] R. Bilder, *On the Search for Extraterrestrial Intelligence (SETI)* *American Journal of International Law* , Volume 114 (1), January 2020, 87 - 95. DOI: <https://doi.org/10.1017/ajil.2019.86>.
- [58] L.I. Tennen, *The Promotion of Space Sustainability Through National Licensing Regimes*, IAC-22, E7.4,x70552, 73rd International Astronautical Congress, Paris, 18-22 September, 2022.
- [59] P. M. Sterns, L. I. Tennen, *Lacuna in the updated planetary protection policy and international law*. *Life Sciences in Space Research* 23 (2019) 10–21.
- [60] K. Denning, *Unpacking the Great Transmission Debate*. *Acta Astronautica* 67 (2010) 1399–1405. <https://doi.org/10.1016/j.actaastro.2010.02.024>.
- [61] M. Michaud, *Contact with Alien Civilisations: Our Hopes and Fears about Encountering Extraterrestrials*, Springer Science & Business Media, 2010.
- [62] D. Vakoch, *Responsibility, capability, and Active SETI: Policy, law, ethics, and communication with extraterrestrial intelligence*. *Acta Astronautica* 68 (2011) 512–519.
1. [63] J. Gertz, *Reviewing METI: A Critical Analysis of the Arguments*, *Journal of the British Interplanetary Society*, Vol. 69, 31-36, 2016; available: <https://arxiv.org/ftp/arxiv/papers/1605/1605.05663.pdf>

[64] P. Hatfield, L. Trueblood, SETI and democracy, *Acta Astronautica*, 202, 180, 2021, 596-603, ISSN 0094-5765, <https://doi.org/10.1016/j.actaastro.2020.11.053>.

[65] K. C. Smith, METI or REGRETTI: Ethics, Risk, and Alien Contact In Social and Conceptual Issues in Astrobiology, K.C. Smith, C. Mariscal (eds.), Oxford University Press. May 2020, 209-236. <https://doi.org/10.1093/oso/9780190915650.003.0013>

[66] C. Haramia, J. DeMarines, The Imperative to Develop an Ethically-Informed METI Analysis, *Theology and Science* 17(1), 2019, 38-48. <https://doi.org/10.1080/14746700.2019.1557800>.

[67] C. Haramia, J. DeMarines, An Ethical Assessment of SETI, METI, and the Value of Our Planetary Home. *Astrobiology: Science, Ethics, and Public Policy*, Chapter 13. Book Editors: O. Alfonso, C. Torres, T. Peters, J. Seckbach, R. Gordon. Wiley. 2021. <https://doi.org/10.1002/9781119711186.ch13>

[68] S. J. Dick, Should We Message ET, and Is an Asilomar Consultation Process Possible?, In S. J. Dick, *Space, Time, and Aliens*. Springer. https://doi.org/10.1007/978-3-030-41614-0_15.

[69] A. Azua-Bustos, J. Benford, et al “Regarding Messaging to Extraterrestrial Intelligence (METI)/Active Searches for Extraterrestrial Intelligence (Active SETI)”, 2015, Available: https://setiathome.berkeley.edu/meti_statement_0.html.

[70] Brian McConnell (2020). The interstellar communication relay. *International Journal of*

Astrobiology 19, 419–422.

<https://doi.org/10.1017/S1473550420000178>

[71] G. Profitiliotis, *Anticipatory Science Diplomacy as a Governance And Cooperation Mechanism for the Search for Extraterrestrial Life Science Policy Journal JSPG*, 2(3) August 2022 www.sciencepolicyjournal.org/uploads/5/4/3/4/5434385/profitiliotis_jspg_20-3.docx.pdf.

[72] S. J. Dick, *Astrobiology, discovery, and societal impact*, Cambridge University Press 2018.

[73] Jill Tarter 2022, “When Do We Get To Drink the Champagne?” in Denning et al, *Post-detection for the 2020s: fundamentals, current concerns, and new discussions*. Breakout session, Penn State SETI Symposium, June 2022.

<https://sites.psu.edu/setisymposium2022/files/2022/06/Post-Detection-breakout-final-program-19-June-2022.pdf>

[74] S. Z. Sheikh, S. Smith, D. C. Price, et al., Analysis of the Breakthrough Listen signal of interest blc1 with a technosignature verification framework. *Nature Astronomy*. 5 (11) (2021) 1153–1162. arXiv:2111.06350. doi:10.1038/s41550-021-01508-8

[75] A.E. Goodman, *Diplomatic and Political Problems Affecting the Formulation and Implementation of an International Protocol for Activities Following The Detection of a Signal from Extraterrestrial Intelligence*, *Acta Astronautica* 21(2) (1990) 106.